

(TR-104) PYTHON WITH AI/ML

Training Day 14 Report

Date: 29 January 2026

Topic Covered: Pandas and Matplotlib – Data Analysis & Data Visualization

The fourteenth day of training focused on performing data analysis using Pandas and data visualization using Matplotlib. The session included working on real-world datasets, handling missing values, grouping data, and creating different types of plots for insights.

- Created and cleaned DataFrames by handling missing values.
- Removed rows with missing client information and filled missing budget values using the average.
- Filtered projects based on budget conditions for analysis.
- Performed grouped analysis of ad spend data based on platform.
- Created bar charts to visualize total ad spend distribution across platforms.
- Generated scatter plots to analyze relationship between ad spend and revenue.
- Identified and annotated the highest revenue point using `plt.annotate()`.
- Created CTR (Click-Through Rate) column and performed time-series analysis.
- Plotted line charts for clicks over time with markers and annotations for peak values.
- Applied sorting and aggregation techniques for better data insights.

Blog Link:

<https://medium.com/@jaspinderkaurwalia855/visualizing-data-with-matplotlib-a-beginners-guide-4d1e70dbfc25>

GitHub Link:

https://github.com/JaspinderKaurWalia26/Python_Practice_Codes/tree/main/29-01-2026